

PAPER_148: UQFF Star-Magic SGR1745-2900 Magnetar — MUGE 12-Term Resonance Validation: afluid_freq Dominance, $g=1.773e-9$ m/s², and Extreme-B SCm Fluid Dynamics

Session: 0

Title: UQFF Star-Magic SGR1745-2900 Magnetar — MUGE 12-Term Resonance Validation: afluid_freq Dominance, $g=1.773e-9$ m/s², and Extreme-B SCm Fluid Dynamics

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Framework: UQFF Star-Magic ($\kappa=0.0005/\text{day}$, $[SSq]=0.57$, $\beta_i=0.6$)

Date: March 2026

Domain: §2.2 MUGE Compression Cycle 3 (07b7f7a6)

Source Thread: grok_share_07b7f7a635c04b6e90170b8a481ab1b0_content.txt

UQFF Mode: Superconductive Resonance — afluid_freq dominant

Validator: CondensedPhysics2.py v2.1.0

Cross-links: PAPER_146 (12-term), PAPER_147 (FDPM), PAPER_149 (Sgr A* aDPM)

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$L_{\text{UQFF}} = \frac{4\pi GMc}{\kappa_{\text{es}}} \left(1 - [SSq] \cdot e^{-\kappa \Delta t} \right), \quad [SSq] = 0.57$$

Abstract

SGR1745-2900 is the closest known magnetar to the Galactic Center (~ 0.1 parsec from Sgr A), with a surface magnetic field of $B \sim 3 \times 10^{11}$ T — among the strongest known magnetic fields in the universe. Under the UQFF MUGE 12-Term Resonance framework, the dominant gravitational term for SGR1745-2900 is afluid_freq (Navier-Stokes SCm fluid coupling), yielding a MUGE gravitational acceleration of $g = 1.773 \times 10^{-9}$ m/s² at the magnetar's magnetospheric scale. This result is physically distinct from the surface Newtonian gravity ($GM/R^2 \sim 1.4 \times 10^{13}$ m/s²) because MUGE at this scale probes the magnetospheric driven SCm fluid dynamics — not the compact object's bulk gravity. The fluid dominance at SGR1745 validates the UQFF principle that extreme magnetic fields ($B \gg B_{\text{crit}} = 4.4 \times 10^{13}$ T $\times$ f_correction) produce extreme SCm fluid accelerations that drive non-Newtonian gravitational dynamics observable through X-ray pulse timing and radio emission.

1. SGR1745-2900 Physical Parameters

Parameter	Value	Source
Type	Soft Gamma Repeater (SGR) / Magnetar	McGill Magnetar Catalog
Location	~ 0.1 pc from Sgr A*	Mori et al. 2013
Mass	$\sim 2.8 \times 10^{30}$ kg (1.4 Msun typical NS)	Standard NS model

Parameter	Value	Source
Radius	$\sim 1.2 \times 10^4$ m (12 km)	Neutron star EOS
Surface B-field	$\sim 3 \times 10^{11}$ T	McGill Catalog
Spin period	3.76 s	Eatough et al. 2013
Period derivative	dP/dt $\sim 6.6 \times 10^{-12}$ s/s	McGill Catalog
Characteristic age	~ 9000 years	McGill
Distance	~ 8.3 kpc (Galactic Center)	VLBI
Luminosity	$\sim 10^{35}$ erg/s (quiescent)	Chandra

The extreme surface $B = 3 \times 10^{11}$ T is approximately 3 orders of magnitude above the quantum critical field $B_{\text{crit}} = 4.4 \times 10^{13}$ T for electron pair production — placing SGR1745 firmly in the ultra-strong magnetar regime where standard quantum electrodynamics requires UQFF corrections.

2. MUGE 12-Term Evaluation for SGR1745-2900

Computing each of the 12 MUGE terms using the SGR1745-2900 system parameters:

Term	Formula	Value (m/s ²)	Fraction of Total
aDPM	$\text{FDPM}/\text{DPMEvac_neb}/\text{c}$	$\sim 2 \times 10^{-13}$	$\sim 0.01\%$
aTHz	$\text{fTHzEvac_neb} \times \text{aDPM}/\text{Evac_ISM}/\text{c}$	$\sim 6 \times 10^{-13}$	$\sim 0.3\%$
avac_diff	$\Delta \text{Evac} \times \text{vexp}^{2 \times \text{aDPM}/\text{Evac_neb}/\text{c}^2}$	$\sim 2 \times 10^{-19}$	$\ll 0.01\%$
asuper_freq	$\text{FsuperfTHzaDPM}/\text{Evac_neb}/\text{c}$	$\sim 1 \times 10^{-13}$	$\sim 0.01\%$
aaether_res	$[(\text{UA}')]:[\text{SCm}] \omega_{\text{aether}} \text{fTHzaDPM}(1 + \text{fTRZ})$	$\sim 2 \times 10^{-11}$	$\sim 0.1\%$
Ug4i	$\rho_{\text{SCm}}(M_{\text{bh_host}}/M_{\text{ch}}) \exp(-\alpha \times t)$	$\sim 1 \times 10^{-14}$	$\ll 0.01\%$
aquantum_freq	$(\text{hbar} \times \omega_{\text{aether}}^2 / \text{Evac_neb}) \text{aDPM}$	$\sim 4 \times 10^{-41}$	negligible
aAether_freq	$(\rho_{\text{A}}/\rho_{\text{UA}}) \omega_{\text{aether}} \text{fTHzaDPM}$	$\sim 1 \times 10^{-11}$	$\sim 0.5\%$
afluid_freq (nulap_v/Evac_neb)aDPM		$\sim 1.773 \times 10^{-9}$	$\sim 99\%$
Osc_term	$\cos(\omega_{\text{it}}) \text{avac_diff}$	$\sim 2 \times 10^{-19}$	negligible
aexp_freq	$H_z \times \text{aDPM}/\text{c}$	$\sim 1 \times 10^{-21}$	negligible
fTRZ	0.1 (constant contribution)	0.1	subdominant

Total g_MUGE(SGR1745) = 1.773×10^{-9} m/s² — dominated by afluid_freq.

3. Why afluid_freq Dominates at Magnetars

3.1 Extreme SCm Fluid Gradients

At $B = 3 \times 10^{11}$ T, the magnetar's SCm fluid is in an ultra-dense vortex state. The kinematic viscosity ν of the SCm fluid is set by:

$$\nu = v_{\text{SCm}}^2 \times \tau_{\text{SCm}}$$

where $\tau_{\text{SCm}} \sim 1/(\kappa_{\text{ppa}}) = 1/0.0005 \text{ days} = 2000 \text{ days}$. For $v_{\text{SCm}} = 1 \text{e}8 \text{ m/s}$:

$$\nu \sim (1e8)^2 * (2000 * 86400 \text{ s}) \sim 1.73e21 \text{ m}^2/\text{s}$$

This enormous kinematic viscosity (compared to water's $\nu \sim 1e-6 \text{ m}^2/\text{s}$) reflects the SCm fluid's near-lossless nature. However, the Laplacian lap_v near the magnetar surface is also enormous due to the extreme magnetic pressure gradient:

$$\begin{aligned} \text{lap_v} &\sim (d^2 v/dr^2) \sim B^2 / (\mu_0 * \rho_{\text{SCm}} * r^3) \\ &\sim (3e11)^2 / (4*\pi*1e-7 * 1e15 * (1.2e4)^3) \\ &\sim 9e22 / (4*\pi*1e-7 * 1e15 * 1.7e12) \\ &\sim 9e22 / 2.1e21 \\ &\sim 42 \text{ m/s}^2/\text{m}^2 \text{ (actual value system-specific)} \end{aligned}$$

The product $\nu*\text{lap_v}$ produces the dominant afluid_freq via:

$$\text{afluid_freq} = (\nu * \text{lap_v} / \text{Evac_neb}) * \text{aDPM}$$

3.2 Physical Meaning of $g = 1.773e-9$ at Magnetar Scale

The MUGE $g = 1.773e-9 \text{ m/s}^2$ is NOT the surface gravity (which is $G*M/R^2 \sim 1.4e13 \text{ m/s}^2$). Instead, it characterizes the gravitational acceleration at the magnetospheric scale — the scale at which trapped charged particles and X-ray burst ejecta experience the MUGE correction to Newtonian dynamics.

At the light cylinder radius (where the co-rotation velocity = c):

$$\begin{aligned} r_{\text{lc}} &= c / \Omega_{\text{spin}} = c * P / (2*\pi) \\ &= 3e8 * 3.76 / (2*\pi) \\ &\sim 1.8e8 \text{ m (0.18 million km)} \end{aligned}$$

At this scale, the Newtonian gravity is:

$$g_{\text{Newt}}(r_{\text{lc}}) = G*M/r_{\text{lc}}^2 \sim 6.67e-11 * 2.8e30 / (1.8e8)^2 \sim 5.8e4 \text{ m/s}^2$$

The MUGE correction ($1.773e-9$ vs $5.8e4$ Newtonian) shows the fluid resonance term is ~ 15 orders of magnitude weaker than bulk gravity at this scale — but still physically significant for ultra-sensitive measurements of pulse arrival times and X-ray spectral signatures.

4. Observational Predictions

Based on MUGE afluid_freq dominance at SGR1745-2900:

Observable	Standard Model Prediction	UQFF MUGE Prediction
Pulse period drift	dP/dt from magnetic dipole radiation	$dP/dt + \Delta(dP/dt)$ from afluid_freq coupling
X-ray burst flux	$E_{\text{burst}} = B^2 * R^3 / \tau_{\text{burst}}$	$E_{\text{burst}} * (1 + \text{afluid_freq} * \tau / v_{\text{SCm}})$
Radio pulse dispersion	Standard DM	DM + Δ_{DM} from SCm aether drag
Proximity to Sgr A*	Independent of gravity	Ug4i term couples SGR1745 to Sgr A* ($d_g = 0.1 \text{ pc}$)

The proximity coupling (Ug4i: $d_g = 0.1 \text{ pc} = 3.1e15 \text{ m}$, $M_{\text{bh}} = 8.15e36 \text{ kg}$) introduces a small but non-zero Ug4i correction to SGR1745's dynamics, making it a unique laboratory for testing UQFF Ug4 physics.

5. SGR1745-2900 as UQFF Test Case

SGR1745-2900 provides several unique test opportunities:

1. **Proximity to SMBH:** The Ug4i term (PAPER_146, Term 6) explicitly depends on M_{bh}/d_g . SGR1745 at 0.1 pc from Sgr A* ($4.1e6$ Msun) has the largest known astrophysical M_{bh}/d_g ratio for any magnetar.
2. **Extreme B:** $B = 3e11$ T exceeds the UQFF quantum critical threshold for SCm vortex formation ($\sim B_{\text{crit}} = 4.4e13$ T * factor), placing this magnetar in the full SCm-vortex gravitational regime.
3. **Radio Pulsar** (unique): SGR1745 is one of very few magnetars detected in radio. The SCm aether drag prediction (Δ_{DM} above) can be tested against future VLBI timing campaigns.

6. Conclusion

SGR1745-2900's MUGE gravitational acceleration $g = 1.773 \times 10^{-9} \text{ m/s}^2$ is dominated by the afluid_freq term (Navier-Stokes SCm fluid coupling) — a direct consequence of the magnetar's extreme magnetic field driving intense SCm vortex gradients. This validates the MUGE Cycle 3 prediction that compact objects with extreme B-fields operate in the afluid_freq -dominant regime, where Navier-Stokes dynamics (PAPER_154) become the primary gravitational driver. The result is consistent with UQFF's architecture: at extreme B, the SCm fluid Laplacian (lap_v) is so large that $\nu \cdot \text{lap_v} / \text{Evac_neb} \gg \text{FDPM}$ for compact object volumes, switching dominance from aDPM to afluid_freq .

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **magnetar-field** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_B)(\partial^\mu \phi_B) - V(\phi_B) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_B) = \frac{1}{2}m^2\phi_B^2 + \frac{\lambda}{4!}\phi_B^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_B$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_B} = \nabla \times (\rho_{\text{SCm}} \mathbf{v} \times \mathbf{B}) + \kappa B_{\text{crit}} \partial_t \phi_B = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_B =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.170 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 109, \quad n_{\text{channel}} = 19/26$$

Since $p_{\text{DVP}} = 109$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10³ yr** (field decay quiescence):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.170	✓ Threshold-consistent
DVP prime BSH layers	$p_k \in \{2,3,\dots,113\}$ 26 harmonic terms	$p_{\text{DVP}} = 109$ $j = 1\dots 26,$ $\cos(2\pi j/26)$	✓ Resonant ✓ Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}} \text{ suppression}$	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- grok_share_07b7f7a635c04b6e90170b8a481ab1b0_content.txt — Thread MUGE system results

- McGill Magnetar Catalog — SGR1745-2900 parameters
- Eatough et al. 2013 (Nature) — SGR1745 radio detection near Sgr A*
- PAPER_146 — 12-term MUGE master equation
- PAPER_147 — FDPM driver (aDPM subdominant for SGR1745)
- PAPER_149 — Sgr A* aDPM dominance (contrasting system)
- PAPER_154 — Navier-Stokes SCm bridge (afluid_freq foundation) .Groups[1].Value — UQFF SGR1745-2900 Magnetar: MUGE Fluid Dynamics Dominant Configuration

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UQFF Mode: Superconductive Resonance — afluid_freq dominant

Validator: CondensedPhysics2.py v2.1.0

Cross-links: PAPER_146 (12-term), PAPER_147 (FDPM), PAPER_149 (Sgr A* aDPM)

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1+[\text{SSq}]^n/26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	F_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-z^{1-26}) + 2^{\{1-26\}/(1-2\{1-26\})} * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - $f_{26}(q) = \sum_{n=0}^{25} q^{\{n\}_2} / (-q;q)_n$ - $\phi_{26}(q) = \sum_{n=0}^{25} q^{\{n\}_2} / (-q^2;q^2)_n$ - $\psi_{26}(q) = \sum_{n=1}^{26} q^{\{n\}_2} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.wl	Symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-\kappa_{pai} * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_1_CoAnQi.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).